DATA 308: Homework 5

Multiple linear regression

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```
library(ISLR2)
library(janitor)
library(tidyverse)

data("Carseats")
```

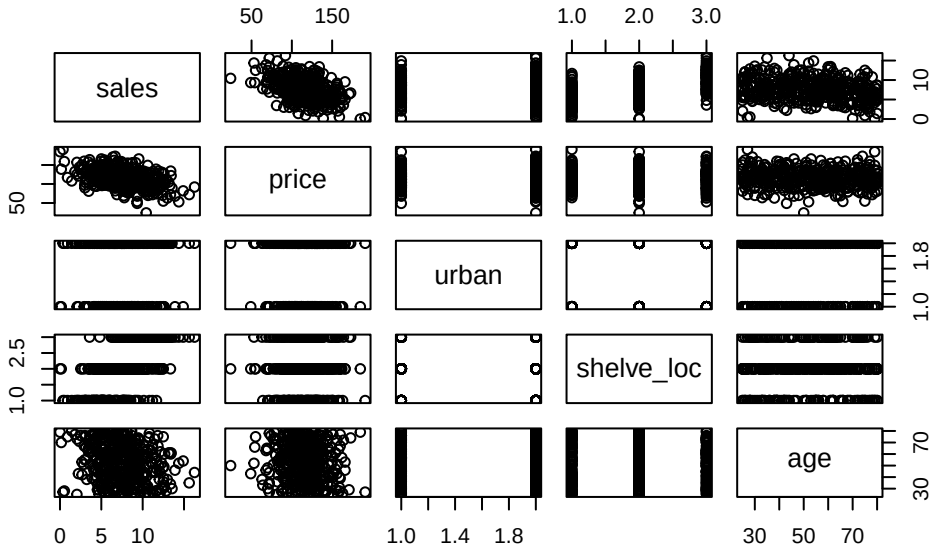
Question 1

This question should be answered using the `car_seats` data set. `car_seats` is a cleaned version of the data I made, to view the data documentation you must search `?Carseats`

```
car_seats <- Carseats |>
  mutate(
    ShelveLoc = factor(ShelveLoc,levels = c("Bad", "Medium", "Good")),
    US = factor(US, levels = c(FALSE, TRUE))
  ) |>
  clean_names()
```

Run the code below to view 2-D scatter plots of sales plotted against selected predictors.

```
car_seats |>
  select(sales, price, urban, shelve_loc, age) |>
  pairs()
```



Question 1a

Fit a multiple regression model to predict `sales` using `price`, `urban`, `shelve_loc` and `age`.

Response

```
car_seat_mlr <- lm(sales ~ price + urban + shelve_loc + age, data = car_seats)

summary(car_seat_mlr)
```

```
Call:
lm(formula = sales ~ price + urban + shelve_loc + age, data = car_seats)

Residuals:
    Min      1Q  Median      3Q     Max
-4.7390 -1.1662  0.0192  1.1601  5.1039

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     14.790575   0.570992  25.903   <2e-16 ***
```

```
price            -0.060588   0.003700 -16.376   <2e-16 ***
urbanYes          0.359849   0.191329   1.881   0.0607 .
shelve_locMedium  2.002595   0.213466   9.381   <2e-16 ***
shelve_locGood    4.974159   0.259650  19.157   <2e-16 ***
age              -0.050366   0.005405  -9.319   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.736 on 394 degrees of freedom
Multiple R-squared:  0.627, Adjusted R-squared:  0.6223
F-statistic: 132.5 on 5 and 394 DF,  p-value: < 2.2e-16
```

Question 1b

Write out the theoretical population model, being careful to handle the qualitative variables properly.

Response

$\text{Sales}_i = \beta_0 + \beta_1 \times \text{Price}_i + \beta_2 \times (\text{Urban}_i = \text{Yes}) + \beta_3 \times (\text{Shelf Location}_i = \text{Medium}) + \beta_4 \times (\text{Shelf Location}_i = \text{Good}) + \beta_5 \times \text{Age}_i + \epsilon_i$

Question 1c

Write out your fitted model, being careful to handle the qualitative variables properly.

Response

$\widehat{\text{Sales}}_i = 14.790575 + (-0.060588 \times \text{Price}_i) + I(0.359849 \times \text{Urban}_i = \text{Yes}) + I(2.002595 \times \text{ShelveLoc}_i = \text{Medium}) + I(4.974159 \times \text{ShelveLoc}_i = \text{Good}) + (-0.050366 \times \text{Age}_i)$

Question 1d

Provide an interpretation of each coefficient in the model. Be careful—some of the variables in the model are qualitative!

Response

For `sales` in thousands, we estimate:
- A decrease of $60.59 for each unit increase in `price`.
- An increase of $359.85 for stores in `urban` locations.
- An increase of $2002.60 on car seats with a medium quality shelving location in stores.
- An increase of $4974.60 on car seats with a good quality shelving location in stores.
- A decrease of $50.37 for every unit increase of average `age` of local population.

(Every β was multiplied by 1000 because sales is given in thousands)

Question 1e

Using the original full model, obtain 95 % confidence intervals for each of the coefficients (i.e., slopes).

Response

```
confint(car_seat_mlr)
```

```
                      2.5 %      97.5 %
(Intercept)      13.66800360 15.91314707
price            -0.06786160 -0.05331375
urbanYes         -0.01630394  0.73600216
shelve_locMedium  1.58292044  2.42226961
shelve_locGood    4.46368643  5.48463105
age              -0.06099163 -0.03973973
```

Question 1f

If we wanted confidence intervals with a confidence level of 99%, would they be narrower or wider than the 95% confidence intervals?

Response

They would be wider. To increase the possibility of the confidence interval containing the true value, the interval must be wider. A higher confidence interval is a less precise estimate, but more likely to contain the true value.

Question 1g

Using the original full model, compute a 95% prediction interval for the number of sales when a car seat price is $100, its sold in an urban location, has a good shelving location, and the average age of the local population is 30.

Response

```
new_car_seat_data <- data.frame(
  price = 100,
  urban = "Yes",
  shelve_loc = "Good",
  age = 30
)

predict(car_seat_mlr, newdata = new_car_seat_data, interval = "prediction")
```

```
       fit      lwr      upr
1 12.55485 9.107908 16.00178
```

We predict that `sales` for stores with a car seat priced at $100 that are in an urban location, have "Good" shelving location, and the average age of the population is 30 years old to be $12554.85, between $9107.91 and $16001.78 with 95% confidence.

Question 1h

We would expect price to be very related to the number of sales made. Conduct a hypothesis test to assess if any of the remaining explanatory variables are linearly related to number of car seats sold, when price is in the model.

Response

$H_0 : \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$
$H_a :$ At least one of $\beta_2, \beta_3, \beta_4,$ or $\beta_5 \neq 0$

```
# redundant model "car_seats_2" below
car_seats_2 <- lm(
  sales ~ price + urban + shelve_loc + age,
  data = car_seats
)
```

```r
car_seats_price <- lm(
  sales ~ price,
  data = car_seats
)

anova(car_seats_2, car_seats_price)
```

```
Analysis of Variance Table

Model 1: sales ~ price + urban + shelve_loc + age
Model 2: sales ~ price
  Res.Df    RSS Df Sum of Sq     F    Pr(>F)
1    394 1186.9
2    398 2552.2 -4   -1365.3 113.3 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Reject H_0, we found very strong evidence ($p = 2.2 \times 10^{-16}$) that at least one of the explanatory variables, `urban`, `shelve_loc`, or `age`, is linearly related to `sales`, when accounting for `price`.

Question 1i

In which of the models (a) or (subsetted model from part h) is the standard error for the price slope larger? Why do you think that is?

Response

```r
summary(car_seat_mlr)
```

```
Call:
lm(formula = sales ~ price + urban + shelve_loc + age, data = car_seats)

Residuals:
    Min      1Q  Median      3Q     Max
-4.7390 -1.1662  0.0192  1.1601  5.1039

Coefficients:
```

```
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)      14.790575   0.570992  25.903   <2e-16 ***
price            -0.060588   0.003700 -16.376   <2e-16 ***
urbanYes          0.359849   0.191329   1.881   0.0607 .
shelve_locMedium  2.002595   0.213466   9.381   <2e-16 ***
shelve_locGood    4.974159   0.259650  19.157   <2e-16 ***
age              -0.050366   0.005405  -9.319   <2e-16 ***
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Residual standard error: 1.736 on 394 degrees of freedom
Multiple R-squared:  0.627, Adjusted R-squared:  0.6223
F-statistic: 132.5 on 5 and 394 DF,  p-value: < 2.2e-16
```

```
summary(car_seats_price)
```

```
Call:
lm(formula = sales ~ price, data = car_seats)

Residuals:
    Min      1Q  Median      3Q     Max
-6.5224 -1.8442 -0.1459  1.6503  7.5108

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 13.641915   0.632812  21.558   <2e-16 ***
price       -0.053073   0.005354  -9.912   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.532 on 398 degrees of freedom
Multiple R-squared:  0.198, Adjusted R-squared:  0.196
F-statistic: 98.25 on 1 and 398 DF,  p-value: < 2.2e-16
```

The standard error for the `price` slope is smaller in the full model (a), at 0.003700, compared to 0.005354 in the subsetted model (h). One possible explanation is that `price` shares a relationship with other predictor variables in the full model, which may help account for additional variation in `sales` and improve the precision of the estimated `price` slope.

Question 2

In this problem we will investigate the t-statistic for the null hypothesis $H_0 : \beta = 0$ in simple linear regression without an intercept. To begin, run this code to simulate an explanatory x and a response y as follows.

```
set.seed(1) # Setting a random seed makes it reproducible. It is useful when you are going to

x <- rnorm(100) # Generated 100 standard normal random variables
y <- 2 * x + rnorm(100) # Calculates Y with more added randomness

simulated_data <- data.frame(x = x, y = y)
```

Question 2a

Perform a simple linear regression of y onto x from `simulated_data`, without an intercept. Report:

- the coefficient estimate $\hat{\beta}_1$,
- the standard error of this coefficient estimate, and
- the t-statistic and p-value associated with the null hypothesis $H_0 : \beta_1 = 0$.

(You can perform regression without an intercept using the command `lm(y  0 + x, data = simulated_data)`.)

Response

```
simulated_model <- lm(y ~ 0 + x, data = simulated_data)

summary(simulated_model)
```

```
Call:
lm(formula = y ~ 0 + x, data = simulated_data)

Residuals:
    Min      1Q  Median      3Q     Max
-1.9154 -0.6472 -0.1771  0.5056  2.3109

Coefficients:
  Estimate Std. Error t value Pr(>|t|)
```

```
x   1.9939     0.1065   18.73   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9586 on 99 degrees of freedom
Multiple R-squared:  0.7798,    Adjusted R-squared:  0.7776
F-statistic: 350.7 on 1 and 99 DF,  p-value: < 2.2e-16
```

Reported values below and APA style report on last line in LaTeX:
$\beta_1 = 1.9939, \beta_1 \neq 0$
Standard Error $= 0.1065$
Reject $H_0, t(99) = 18.73, p = 2 \times 10^{-16}$

Question 2b

Now perform a simple linear regression of x onto y without an intercept, and report:

- coefficient estimate $\hat{\beta}_1$,
- its standard error, and
- the corresponding t-statistic and p-values associated with the null hypothesis $H_0 : \beta_1 = 0$.

Response

```r
simulated_model_2 <- lm(x ~ 0 + y, data = simulated_data)

summary(simulated_model_2)
```

```
Call:
lm(formula = x ~ 0 + y, data = simulated_data)

Residuals:
    Min      1Q  Median      3Q     Max
-0.8699 -0.2368  0.1030  0.2858  0.8938

Coefficients:
  Estimate Std. Error t value Pr(>|t|)
y  0.39111    0.02089   18.73   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.4246 on 99 degrees of freedom
Multiple R-squared:  0.7798,    Adjusted R-squared:  0.7776
F-statistic: 350.7 on 1 and 99 DF,  p-value: < 2.2e-16
```

Question 2c

What is the relationship between the results obtained in (2a) and (2b)? Specifically, what is the same and what is different, if anything?

Response

Comparing the results from 2a and 2b, only the coefficient estimate and standard error changed among the statistics we were asked to report. The coefficient estimate decreased from 1.9939 to 0.39111, while the standard error decreased from 0.1065 to 0.02089.

The t-statistic remained the same at 18.73 with 99 degrees of freedom, and both models produced a p-value less than (2 $\times$10^{-16}).